Power-Law Initialization for Budget-Constrained Physics-Informed Extreme Learning Machines

Quang Nguyen[a,b,*]

[a]*Department of Physics, International University, Ho Chi Minh City, Vietnam*
[b]*Viet Nam National University Ho Chi Minh City, Vietnam*

Abstract

Physics-Informed Extreme Learning Machines (PIELMs) solve linear PDEs in a single least-squares step, but their accuracy depends critically on the weight scale hyperparameter R_m. We propose power-law weight initialization, drawing weight magnitudes from $P(|w|) \propto |w|^{-\alpha}$, for sinusoidal-basis PIELMs. An ablation identifies the optimal range $\alpha \approx 0.75$–1.25, which spreads neurons across frequency decades rather than concentrating them in a narrow band. Through experiments on eight 1D and three 2D PDE benchmarks, we show: (i) at tight neuron budgets ($h \leq 40$), power-law initialization outperforms Gaussian and uniform by up to $300\times$ in 1D; (ii) the advantage *amplifies* in 2D, exceeding $10^6\times$ at $h = 400$; (iii) power-law degrades more gracefully when R_m is misspecified. We recommend $\alpha = 1$ (log-uniform) as the default. All training uses PDE residuals and boundary conditions only.

Keywords: Physics-Informed Extreme Learning Machines, Power-Law Initialization, Sinusoidal Activation, Spectral Basis Diversity

1. Introduction

Physics-Informed Neural Networks (PINNs) [1] embed PDE residuals into the training loss, enabling mesh-free solvers. However, PINNs require thousands of gradient-descent iterations and suffer from spectral bias—a preference for low-frequency components [2, 3].

Extreme Learning Machines (ELMs) [4] bypass iterative training: hidden weights are frozen at random and only output weights β are solved analytically. Dwivedi and Srinivasan [5] extended this to PDEs as Physics-Informed ELMs (PIELMs), solving a single linear system that enforces the PDE residual and boundary conditions.

A central limitation is sensitivity to the weight scale R_m. Dong [6] showed that accuracy varies by orders of magnitude with R_m and proposed differential evolution to find the optimum—an expensive search that partially negates the one-shot advantage. This sensitivity arises because standard initializations (Gaussian, uniform) concentrate weights in a narrow frequency band.

*Corresponding author
Email address: nquang@hcmiu.edu.vn (Quang Nguyen)

We address this through two ideas: (1) **sinusoidal activation** for PIELM, following recent Fourier-based work [7, 8, 9], making each neuron an explicit Fourier mode; and (2) **power-law weight initialization** $P(|w|) \propto |w|^{-\alpha}$ with $\alpha \approx 1$, which spreads neurons across frequency decades. Tancik et al. [10] explored log-uniform sampling ($\alpha = 1$) for deep Fourier feature networks and concluded that distribution shape does not matter. We show this conclusion does *not* transfer to PIELMs: without iterative training to compensate, distribution shape becomes critical.

Our main findings are:

- At tight neuron budgets ($h \leq 40$), power-law outperforms Gaussian by 10–300$\times$ in 1D (Section 4.2).
- The advantage *amplifies* in 2D: Gaussian fails entirely ($L_2 > 0.6$) at $h = 400$ while power-law achieves 10^{-7}, exceeding $10^6\times$ advantage (Section 4.5).
- The optimal exponent is $\alpha \approx 0.75$–1.25; we recommend $\alpha = 1$ (log-uniform) as the default (Section 4.3).
- Power-law degrades gracefully when R_m is misspecified (Section 4.4).

2. Methods

2.1. PIELM with Sinusoidal Basis

Given a linear PDE $\mathcal{L}[u](x) = f(x)$ on Ω with Dirichlet conditions $u|_{\partial\Omega} = g$, PIELM approximates $\hat{u}(x) = \sum_{j=1}^{h} \beta_j \sin(w_j x + b_j)$, where $\{w_j, b_j\}$ are frozen and only β is solved for. Because $\sin(\cdot)$ is an eigenfunction of differentiation, PDE operators have closed forms:

$$\text{Poisson } (-u'' = f): \quad w_j^2 \sin(w_j x + b_j), \tag{1}$$

$$\text{Helmholtz } (-u'' - k^2 u = f): \quad (w_j^2 - k^2) \sin(w_j x + b_j). \tag{2}$$

For d-dimensional inputs with $\phi_j(\mathbf{x}) = \sin(\mathbf{w}_j^\top \mathbf{x} + b_j)$, the Laplacian gives $\|\mathbf{w}_j\|^2 \cdot \phi_j(\mathbf{x})$, so the formulation extends directly to 2D.

The PIELM system stacks PDE and boundary rows:

$$\begin{bmatrix} A_{\text{pde}} \\ \lambda_{\text{bc}} A_{\text{bc}} \end{bmatrix} \beta = \begin{bmatrix} f_{\text{int}} \\ \lambda_{\text{bc}} g_{\text{bc}} \end{bmatrix}, \tag{3}$$

solved via $\beta = (A^\top A + \lambda I)^{-1} A^\top \mathbf{r}$ in one shot with cost $O(Nh^2 + h^3)$.

2.2. Power-Law Initialization

Standard initializations concentrate weights at a single scale: Gaussian $\mathcal{N}(0, R_m)$ clusters near zero; uniform $\mathcal{U}(-R_m, R_m)$ over-represents frequencies near R_m. We propose drawing weight magnitudes from a truncated power-law distribution:

$$P(|w|) \propto |w|^{-\alpha}, \quad |w| \in [1, R_m], \tag{4}$$

with random sign $\mathrm{sgn}(w_j) \in \{-1, +1\}$ and bias $b_j \sim \mathcal{U}(-\pi, \pi)$. The exponent α controls the frequency allocation: $\alpha = 1$ gives log-uniform (equal probability per decade), $\alpha < 1$ favors higher frequencies, and $\alpha > 1$ favors lower frequencies. Our ablation (Section 4.3) identifies $\alpha \approx 0.75$–1.25 as optimal; we recommend $\alpha = 1$ as the default. In the language of power-law distributions [11], $\alpha = 1$ is the boundary between light-tailed and heavy-tailed regimes.

For sinusoidal activation, each neuron is a random Fourier feature [12, 10] whose frequency is $|w_j|$. Gaussian weights yield a narrowband kernel; power-law weights yield a broadband spectrum—a natural match for physical systems with multi-scale energy content.

3. Experimental Setup

All experiments use Eq. 3 with $\lambda_{\text{bc}} = 100$, $\lambda = 10^{-10}$, double precision, and 20 seeds. 1D benchmarks use $N = 200$ interior points; 2D benchmarks use a 30×30 grid. The reference solution is used only for evaluation.

1D Benchmarks.. (1) Poisson: $u = \sin(\pi x)$; (2) Multi-frequency Poisson: $u = \sin(\pi x) + 0.3\sin(3\pi x) + 0.1\sin(7\pi x)$; (3) Helmholtz ($k$=10): $u = \sin(\pi x)e^{-4x^2}$; (4) High-frequency Poisson: $u = \sin(10\pi x)$; (5) Advection–diffusion (ε=0.02). Additionally: (6) Helmholtz k=20; (7) $u = x(1-x^2)\sin(5\pi x)$; (8) $u = \sin(\pi x)\cos(3\pi x)$. All on $[-1, 1]$ with $u(\pm 1) = 0$.

2D Benchmarks. on $[-1, 1]^2$ with $u|_{\partial\Omega} = 0$: (9) Poisson: $u = \sin(\pi x)\sin(\pi y)$; (10) Multi-frequency: $u = \sin(\pi x)\sin(2\pi y) + 0.3\sin(3\pi x)\sin(\pi y)$; (11) Helmholtz ($k$=5): $u = \sin(\pi x)\sin(\pi y)$.

Initializations.. Power-Law (proposed): $P(|w|) \propto |w|^{-\alpha}$ on $[1, R_m]$, $\alpha = 1$ (default), $b_j \sim \mathcal{U}(-\pi, \pi)$. Gaussian: $w_j \sim \mathcal{N}(0, R_m)$, $b_j \sim \mathcal{N}(0, R_m)$. Uniform: $w_j \sim \mathcal{U}(-R_m, R_m)$, $b_j \sim \mathcal{U}(-R_m, R_m)$. Default: $R_m = 30$, $h = 400$.

4. Results

4.1. 1D Benchmark at Fixed Scale

Table 1 reports results at $h = 400$, $R_m = 30$. All initializations achieve $< 10^{-7}$ on the first three problems. Gaussian wins on high-frequency Poisson and advection–diffusion because $\mathcal{N}(0, 30)$ concentrates mass near the target frequency ($10\pi \approx 31.4$). This motivates the question: when the budget is *tight*, does power-law's diversity become an advantage?

4.2. Neuron Budget Experiment

We sweep $h \in \{20, 40, 60, 80, 100, 150, 200, 300, 400\}$ at fixed $R_m = 30$. Figure 1 shows the results.

At $h = 20$, power-law achieves $L_2 = 2.9 \times 10^{-3}$ on Helmholtz while Gaussian fails (8.9×10^{-1})—a $300\times$ advantage. The mechanism: with 20 neurons, $\mathcal{N}(0, 30)$ produces many weights near ± 30, creating nearly identical basis functions. Power-law distributes weights across $[1, 30]$ in log-space, ensuring each neuron covers a distinct frequency. As h increases beyond 60, all methods converge.

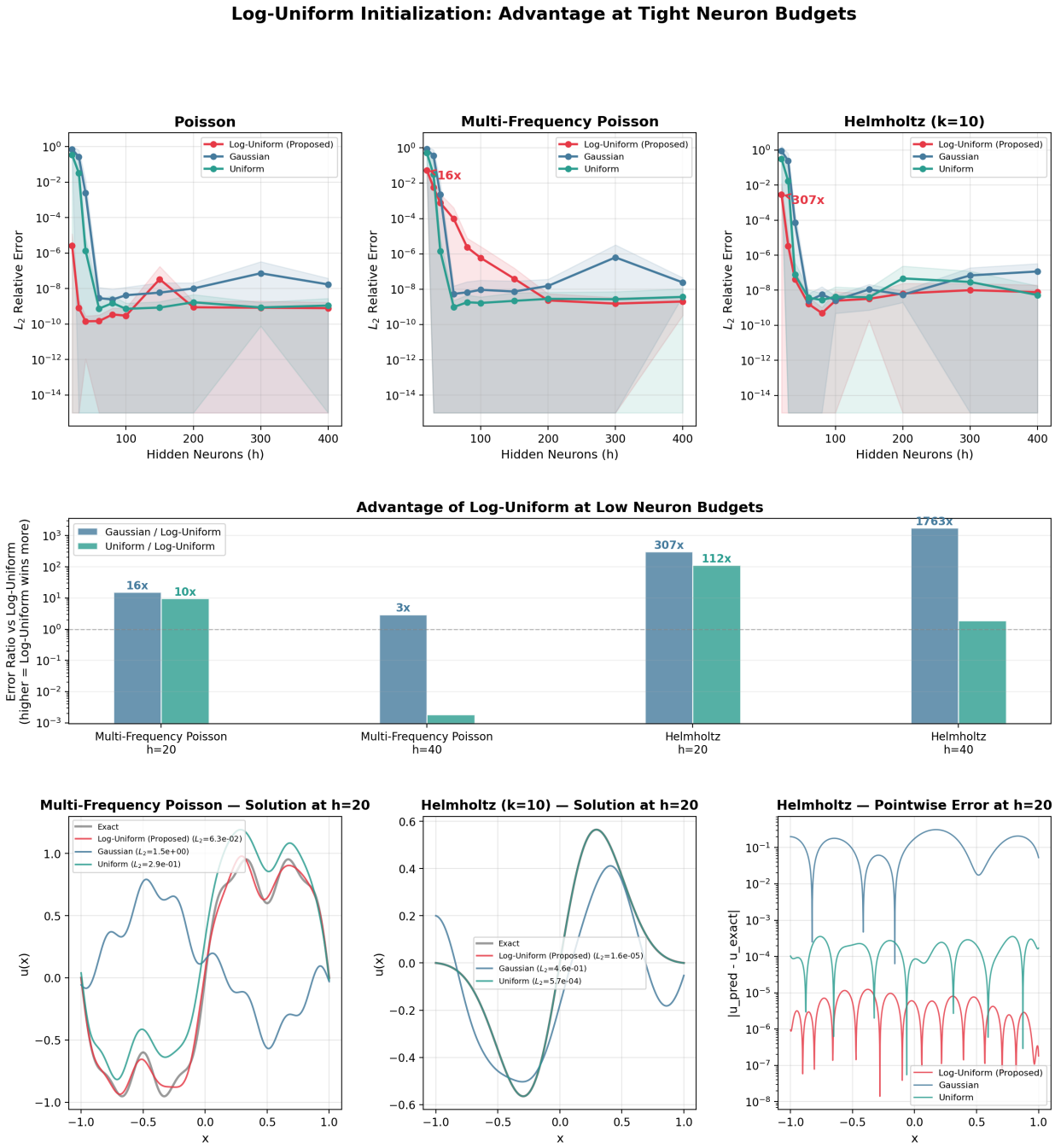

Figure 1: **Power-law initialization at tight neuron budgets.** Top: L_2 error vs h (shaded: ± 1 std). Middle: Gaussian/power-law error ratio at $h = 20$ and 40. Bottom: solution fields at $h = 20$; Gaussian and uniform fail while power-law tracks the exact solution.

Table 1: 1D PIELM results (mean ± std, 20 seeds, h=400, R_m=30). Bold = best.

Problem	**Init**	**Mean L_2 Error**
Poisson	Power-Law	$\mathbf{7.9 \times 10^{-10}}$
	Gaussian	1.7×10^{-8}
	Uniform	1.1×10^{-9}
Multi-Freq	Power-Law	$\mathbf{2.0 \times 10^{-9}}$
	Gaussian	2.4×10^{-8}
	Uniform	3.7×10^{-9}
Helmholtz	Power-Law	7.7×10^{-9}
	Gaussian	1.2×10^{-7}
	Uniform	$\mathbf{5.3 \times 10^{-9}}$
High-Freq	Power-Law	3.0×10^{-5}
	Gaussian	$\mathbf{2.8 \times 10^{-7}}$
	Uniform	6.2×10^{-6}
Adv-Diff	Power-Law	9.6×10^{-4}
	Gaussian	$\mathbf{5.5 \times 10^{-6}}$
	Uniform	1.3×10^{-2}

4.3. Power-Law Exponent Ablation

We sweep $\alpha \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0, 3.0, 4.0\}$ at $h = 400$, $R_m = 30$. For single-frequency Poisson, α has little effect. For multi-frequency problems, accuracy spans eight orders of magnitude across the tested range: from 1.4×10^{-9} at $\alpha = 0.75$ to 6.8×10^{-2} at $\alpha = 2.0$ on multi-frequency Poisson. The optimal range $\alpha \approx 0.75$–1.25 corresponds to near-uniform allocation in log-space, balancing coverage across all frequency decades.

4.4. Scale Robustness

We sweep $R_m \in \{1, 2, \ldots, 150\}$ independently for each initialization (Figure 2).

At optimal R_m, errors are comparable across initializations. However, degradation curves differ: at $R_m = 100$ on multi-frequency Poisson, Gaussian error $= 5.4 \times 10^{-4}$, power-law error $= 7.7 \times 10^{-8}$—a 7,000× gap. Power-law provides a safe default when R_m cannot be tuned per problem.

4.5. Extension to Two Dimensions

Table 2 reports 2D results at $h = 400$, $R_m = 30$.

The advantage is dramatically larger than in 1D: at $h = 400$—where 1D methods converge regardless of initialization—Gaussian and uniform *fail completely* ($L_2 > 0.6$) on all three 2D benchmarks, while power-law achieves 10^{-4} to 10^{-7}.

Figure 3 shows the budget sweep. On Helmholtz 2D, Gaussian never drops below $L_2 = 0.96$ even at $h = 400$, while power-law reaches 6.8×10^{-7}—exceeding $10^6\times$.

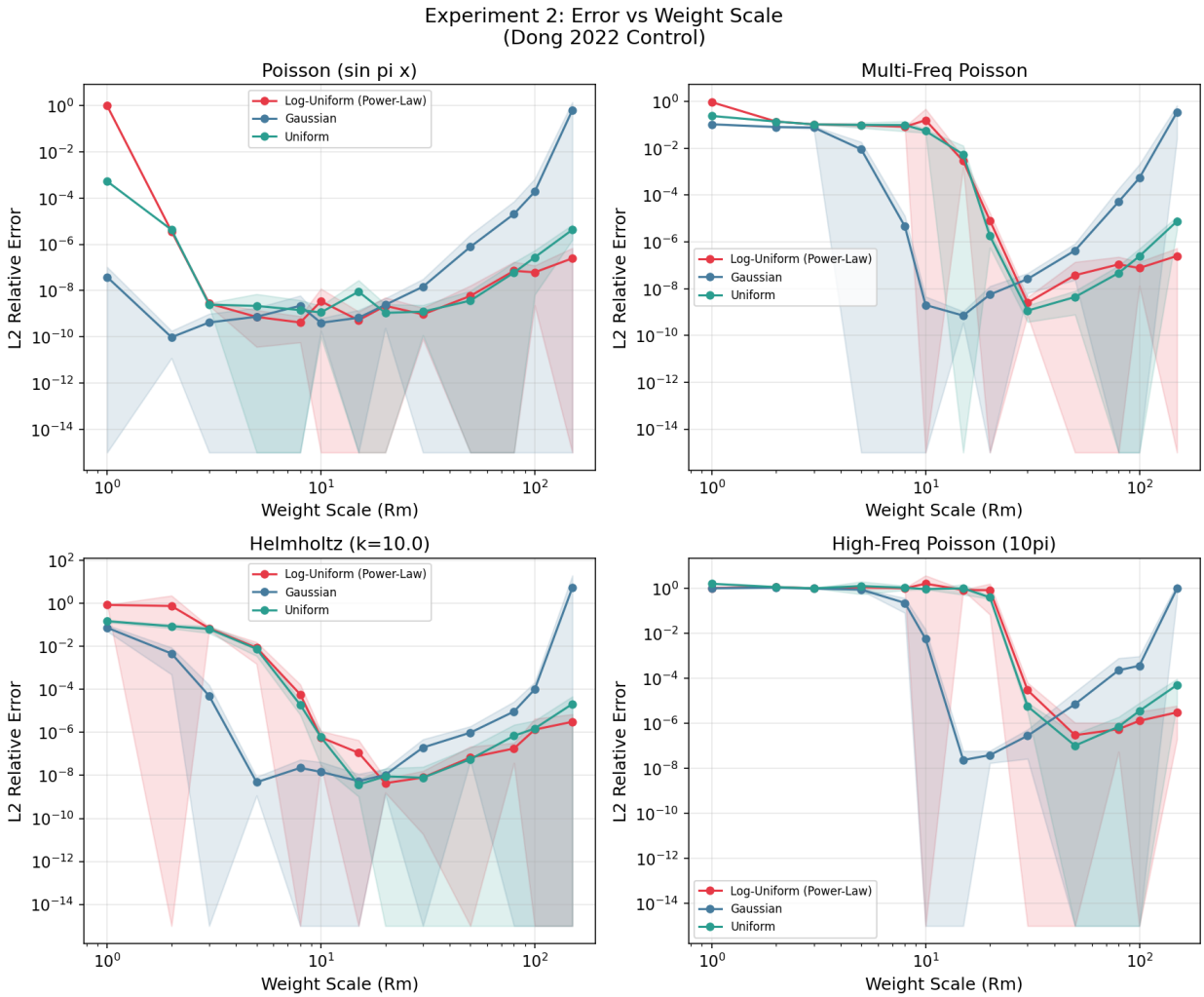

Figure 2: L_2 error vs weight scale R_m (shaded: ± 1 std, 10 seeds). Gaussian degrades sharply when R_m is over-specified; power-law remains robust.

Table 2: 2D PIELM results (mean, 20 seeds, h=400, R_m=30). Bold = best.

Problem	**Power-Law**	**Gaussian**	**Uniform**
Poisson 2D	$\mathbf{2.1 \times 10^{-7}}$	7.8×10^{-1}	6.0×10^{-1}
Multi-Freq 2D	$\mathbf{1.3 \times 10^{-4}}$	6.7×10^{-1}	4.7×10^{-1}
Helmholtz 2D	$\mathbf{8.9 \times 10^{-7}}$	9.5×10^{-1}	8.3×10^{-1}

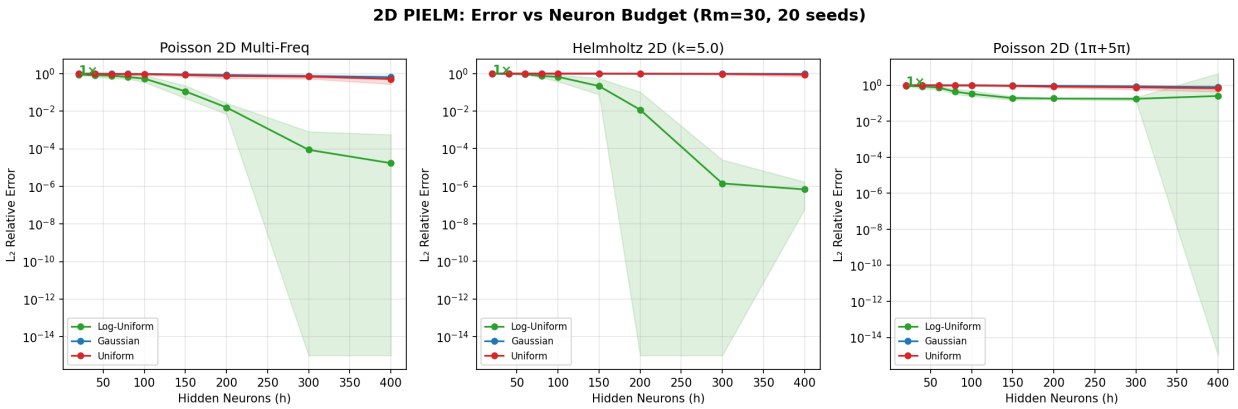

Figure 3: 2D PIELM: L_2 error vs h (R_m=30, 20 seeds). Gaussian and uniform fail even at $h = 400$; the advantage grows with h rather than diminishing.

The mechanism is the same but amplified: Gaussian draws both components from $\mathcal{N}(0, R_m)$, so $\|\mathbf{w}_j\| \approx \sqrt{2}\, R_m$ is concentrated in a narrow band. Power-law distributes magnitudes across decades. In 2D, the waste from duplicate frequencies is more damaging—hence the larger advantage.

5. Discussion

Our results delineate two regimes. When the system is over-parameterized ($h \gg$ required), all initializations achieve comparable accuracy and Dong's [6] finding that scale matters more than distribution holds. When the budget is constrained ($h \leq 60$ in 1D, $h \leq 400$ in 2D), distribution shape becomes critical: power-law's frequency diversity produces a well-conditioned system, while Gaussian wastes capacity on redundant features. This is the regime where Tancik et al.'s [10] conclusion—that only the standard deviation matters—breaks down, because there is no iterative training to compensate for a poor basis.

The 2D results reveal that this regime boundary shifts upward with dimensionality. Problems that are well-resolved at $h = 60$ in 1D require $h > 400$ in 2D for Gaussian, while power-law still solves them. This suggests the advantage will grow further in 3D.

Limitations.. (1) PIELM requires the PDE operator to be linear in β; nonlinear PDEs need iterative schemes. (2) When the solution contains a single high-frequency mode, Gaussian can be superior by concentrating neurons near the target frequency. (3) Global sinusoidal bases struggle with localized features (boundary layers); domain decomposition [13] may help. (4) Extension to 3D requires further study.

6. Conclusions

We proposed power-law weight initialization for sinusoidal-basis PIELMs. The main findings:

1. At low neuron budgets, power-law outperforms Gaussian by up to $300\times$ in 1D by ensuring every neuron covers a distinct frequency scale.
2. The optimal exponent is $\alpha \approx 0.75$–1.25, with $\alpha = 1$ (log-uniform) recommended as the default.
3. Power-law degrades gracefully when R_m is misspecified, reducing the need for per-problem tuning.
4. In 2D, the advantage *amplifies*: Gaussian and uniform fail entirely at $h = 400$ while power-law achieves 10^{-7}, because 2D frequency space concentrates Gaussian weight magnitudes more tightly.

These results position power-law PIELM as a practical choice for budget-constrained PDE solving. Future work will extend to 3D, nonlinear PDEs via iterative PIELM, and adaptive frequency allocation.

Appendix A. Solution Fields for Diverse Problem Types

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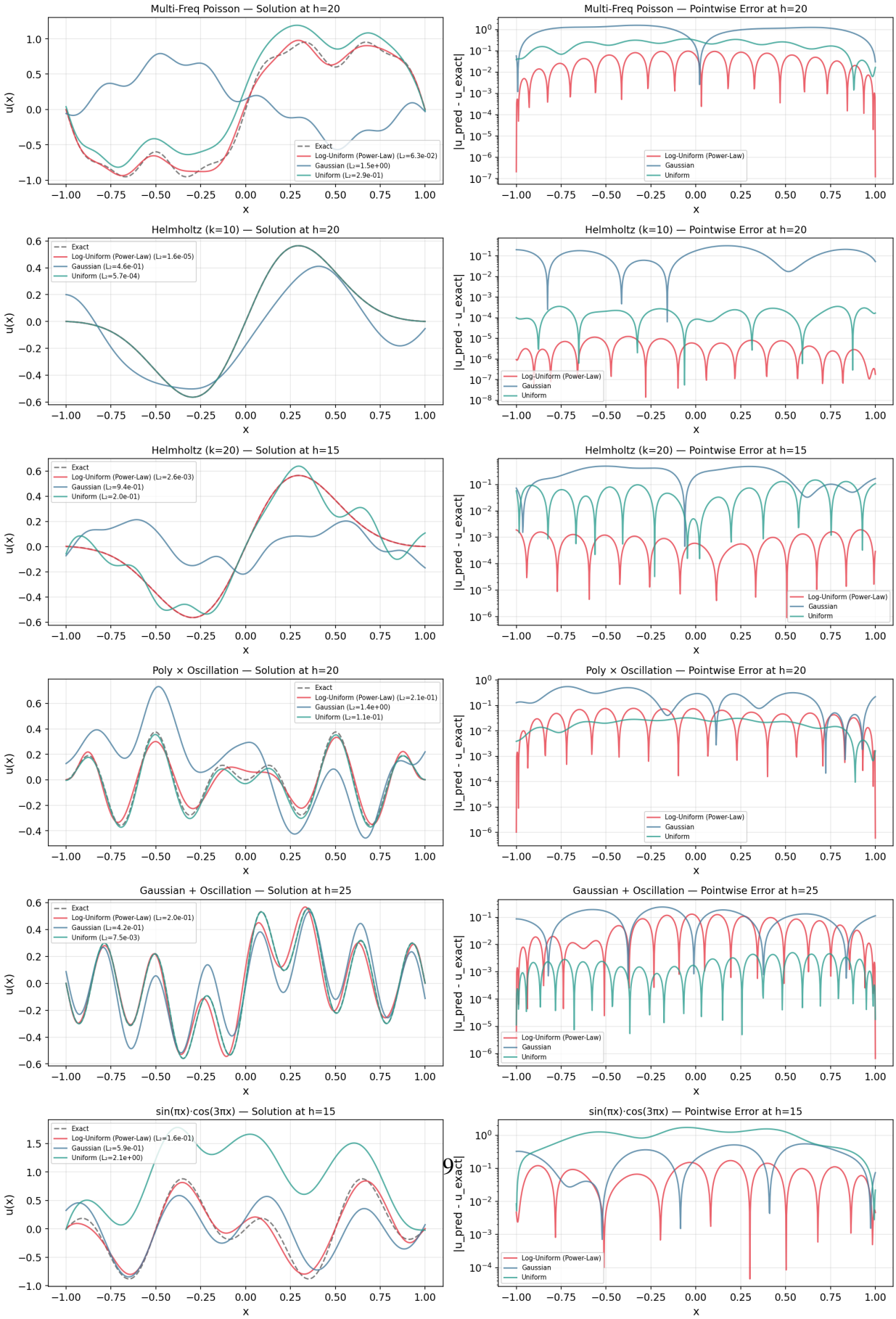
Power-Law Advantage Across Problem Types
(each row at the critical h where power-law works and others fail)



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